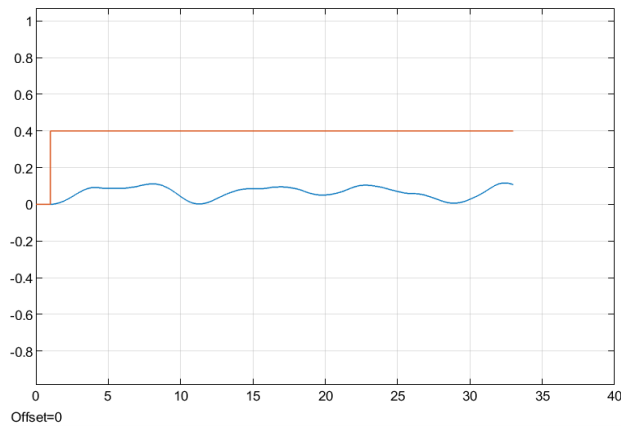
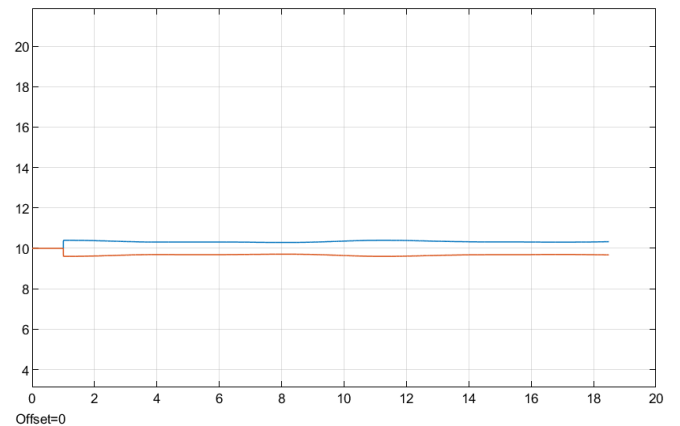






(a) Angle



(b) Motor Voltages

Figure 2: Aero 2 unity feedback step response

3. Add a *proportional* gain block  $k_p$  between the *error sum* blocks as shown in Figure 3.

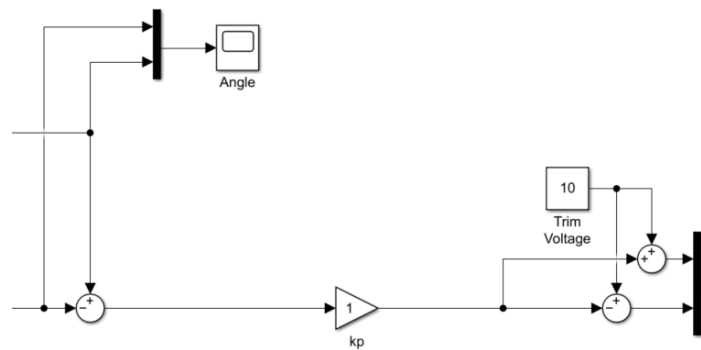


Figure 3: Proportional gain in the feedback loop

4. Vary the proportional gain between 1 and 15. Run the QUARC model with various proportional gains. Take notes on the effect of increasing the proportional gain on the system response. Capture your responses as well.

## Derivative Control

5. Add a *transfer function* and derivative *gain* block  $k_d$  in parallel with the proportional *gain* as shown in Figure 4.

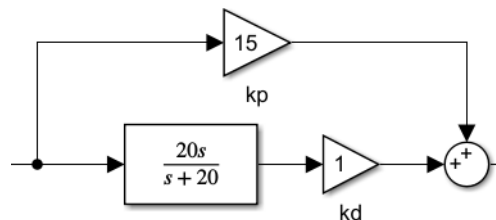


Figure 4: Proportional gain and filtered derivative gain feedback loop

- The angle data from the encoder is a discrete signal, therefore we will have to add some filtering in order to smooth out discontinuities when the encoder count changes. Set the transfer function for the derivative to

$$F(s) = \frac{20s}{s + 20} \quad (1)$$

- Set the proportional gain to 15 and vary the derivative gain between 0 and 10.
- Change the *Input Select* switch to a square wave input.
- Run the QUARC model with various derivative gains. What effects does increasing the derivative gain have on the system response? How does the system response differ when the derivative gain is very small (e.g. <0.25). Capture your responses.
- It is desired that the PD compensated system has a peak pitch that does not exceed  $\pm 0.45$  radians when commanded with a 0.4 radian square wave. Note the minimum value of  $k_d$  which produces a system with acceptable overshoot.

## Integral Control

- The response should now approach the commanded value quickly and settle in a short time. However, even with a large proportional gain, the final position of the Aero 2 still differs from the command value by approximately 0.7 radians. To combat this steady-state error, keeping the proportional and derivative gains at the values identified in the previous step, add an integral gain  $k_i$  as well as an integrator in parallel with the other two gains as shown in Figure 5.

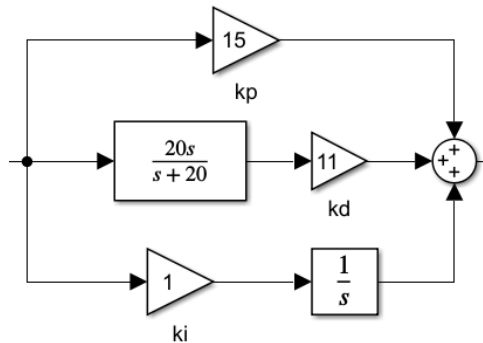


Figure 5: PID gains in a feedback loop

- Vary the *integral gain* between 0 and 10. Take notes on the effect of increasing this gain on the system response, and capture your responses.
- Given the specification that the system settles within 3% of the commanded value, this means that the steady-state value must be in the range of 0.388 to 0.412 radians. Note the integral gain that results in a system that meets this requirement.

## Response Tuning

- If your response does not match the overshoot (<0.45 radians peak pitch) and peak time (<2.2 seconds) specifications from previous steps, try tuning your control gains until your response satisfies all three requirements (including steady-state error). Plot the resulting system response

overlaid on the command waveform and linear representations of the peak overshoot limits. Record the final PID gains.

15. Stop and close the model. Power OFF the Aero 2.